Brett Angeles
1667 Hillsborough Street, Chula Vista, CA 91913
Phone: 619-204-2841 Email: brett.d.angeles@gmail.com

EDUCATION

University of California, San Diego (UCSD)
B.S. Mechanical Engineering - Controls & Robotics Specialization • August 2025

TECHNICAL SKILLS

CAD & Design: SolidWorks, AutoCAD, Autodesk Revit, Fusion 360, Autodesk Inventor
Engineering: Engineering Calculations, Troubleshooting, FEA (SolidWorks Simulation), MATLAB, GD&T
Fabrication: Manual Lathe, Manual Mill, CNC Equipment, Drill Press, Hand & Power Tools, 3D Printing
Software: Microsoft Office, Google Workspace
Languages: English (Native), Korean (Limited Working Proficiency)

WORK EXPERIENCE

Tepexpa — Mechanical Engineer (2025–Present)
Regenerative braking trailer prototype development involving mechanical design, fabrication, and prototype testing

- Designed and developed an electromechanical prototype as part of a 5-person engineering team
- Created SolidWorks models and fabrication drawings for prototype components
- Designed and modified mechanical assemblies to simplify fabrication and improve drivetrain integration
- Performed gear ratio, torque, and energy calculations for drivetrain design analysis
- Fabricated, assembled, and evaluated prototype components using welding and machining
- Conducted braking tests at 15–25 mph using voltage/current measurements for analysis
- Identified design deficiencies and developed engineering solutions during prototype testing

Machine Shop Assistant — *UC San Diego* (2024-2025)
- Interpreted engineering drawings and dimensional tolerances
- Operated manual mills, lathes, CNC equipment, and drill presses for fabrication
- Verified dimensions and inspected fabricated parts for tolerance compliance
- Assisted students with machining operations and safe equipment usage

COSMOS Undergraduate Cluster Assistant — UC San Diego (2023–2024)
- Led hands-on engineering labs involving CAD, 3D printing, and mechanical systems
- Supported technical instruction in statics, kinematics, CAD drafting, and engineering fundamentals
- Coordinated with lead faculty to design a four-week structural & mechanical engineering course for students

PROJECTS

UCSD Concrete Canoe Team — Construction Lead (2024–2025)
- Led a 6 person team responsible for mold fabrication, assembly, and build scheduling
- Designed and fabricated structural mold frameworks and transport systems
- Coordinated tooling, fabrication sequencing, and mold construction activities
- Implemented process improvements to increase build consistency and efficiency

UCSD Concrete Canoe Team — Construction Junior Lead (2023–2024)
- Constructed canoe mold and supporting structures for fabrication
- Designed and built transport systems for handling completed structures
- Achieved 4th place at the 2024 Pacific Southwest Symposium